

LSTM Model To Predict Stock Price

Problem Description

This project focuses on developing an **LSTM-based predictive model** to forecast the target price of a stock using historical data. By analyzing 5-minute interval OHLC (Open, High, Low, Close) data, the model captures short-term price trends and patterns, helping traders optimize their strategies. The primary objective is to predict the next interval's price, providing actionable insights for setting stop-loss and target levels. Using the sequential learning capabilities of LSTM, the model addresses the inherent volatility and non-linear nature of financial markets. This solution empowers traders to make data-driven decisions, improve risk management, and enhance profitability.

DataSet

The Dataset is taken from yahoo finance which consists of historical data of a stock.

Dataset Consists of Open, High, Low, Close prices, and Volume .

Predicted closing price for the next intervals (e.g., Close price).

Data Preprocessing: Converting data into the Heikin-Ashi form then Normalising using MinMaxScaler or StandardScaler, and creating time-series windows for LSTM input.

Data Splitting: 70-80% for training, 30-20% for validation.

Data-Preprocessing

Data fetched from yahoo finance, containing high, low, close, open and volume.

Data converted to the Heikin-Ashi form to smoothen the data and highlight trends more clearly.

Min-Max scaling the data such that stock prices and related features are scaled uniformly, improving the performance, stability, and interpretability of the machine learning model .

$$X_{scaled} = (X - X_{min}) / (X_{max} - X_{min})$$

Implementation

LSTM Layer: Used for capturing time-dependent patterns in sequential data, crucial for time-series prediction tasks.

Dense Layer: Used to output predictions from the LSTM layer hidden states.

Dropout Layer: Applied to prevent overfitting by randomly deactivating neurons during training.

Callbacks: EarlyStopping and ModelCheckpoint are used to monitor validation performance and save the best model during training.

Model Configuration

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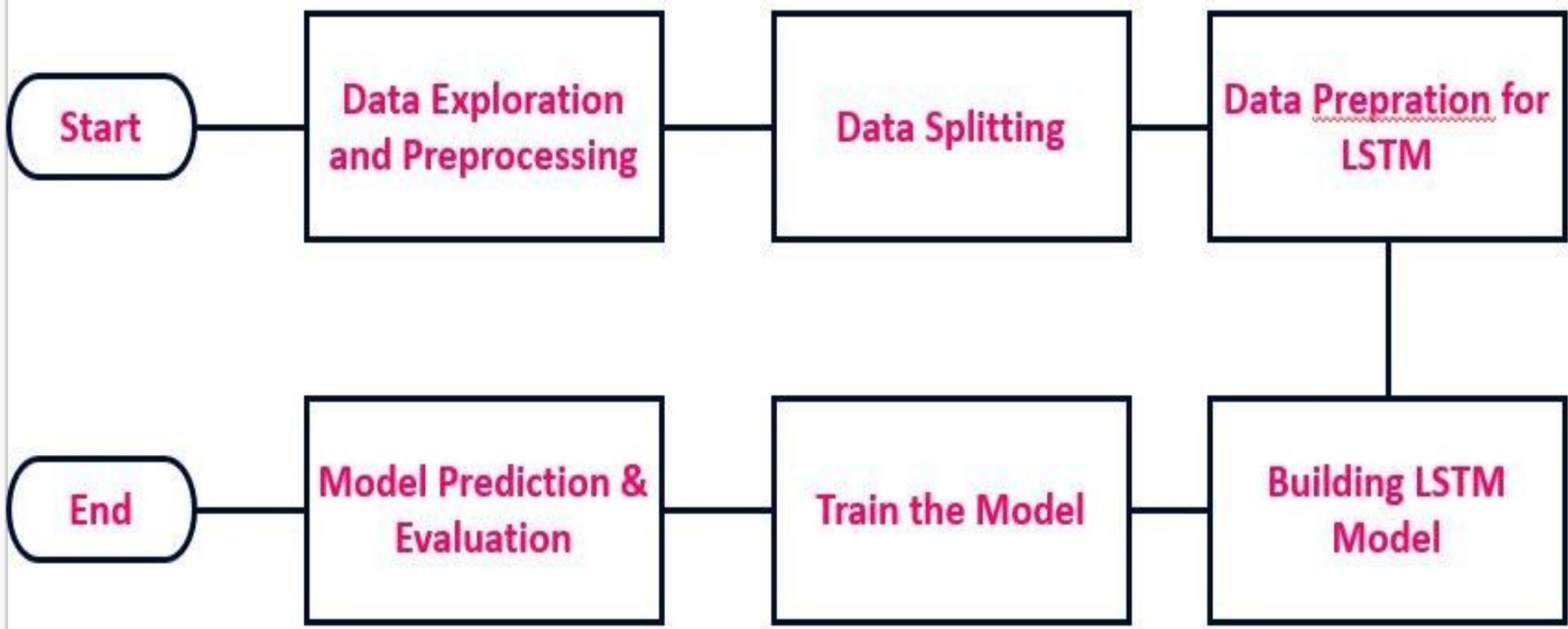
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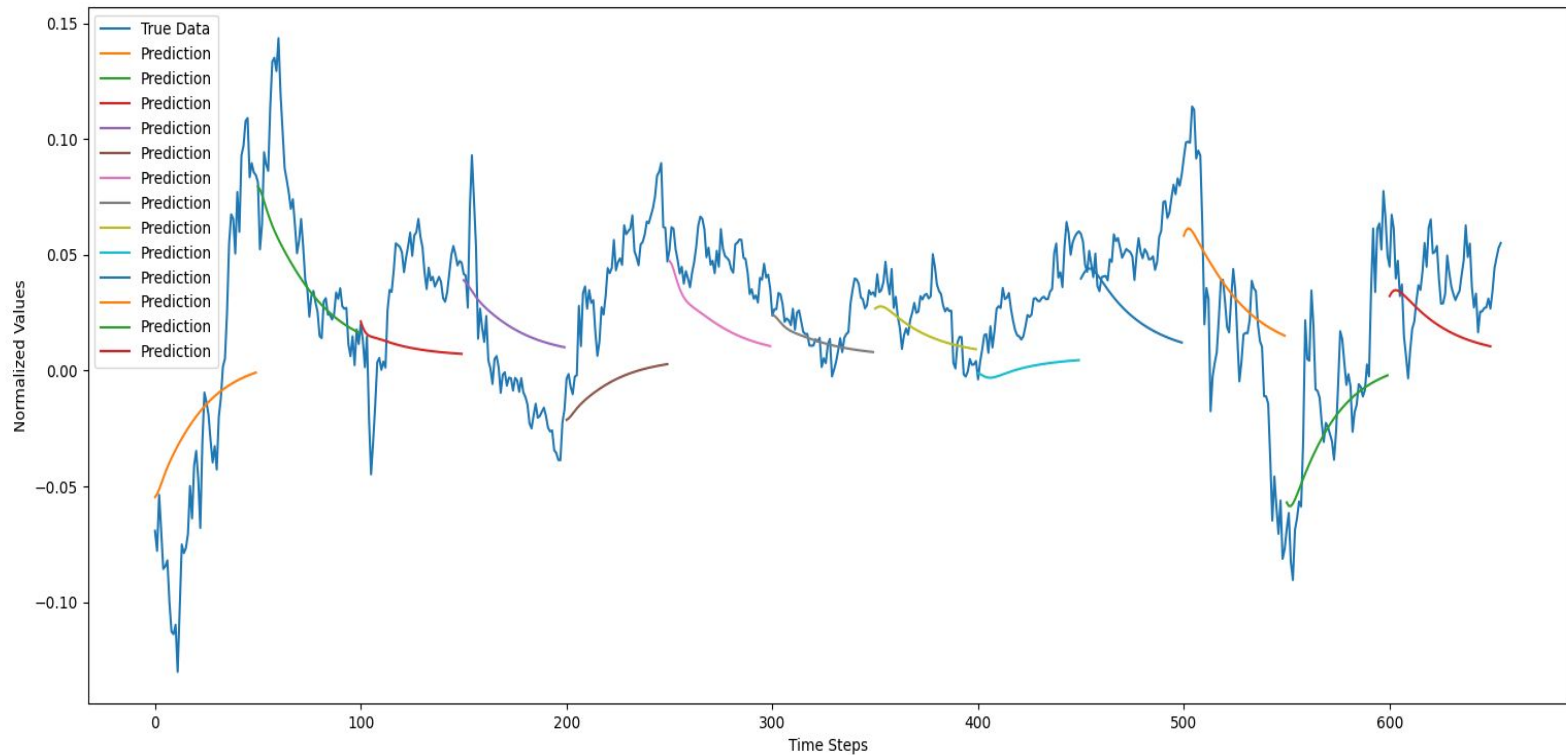
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Prediction chart for 10 years of chart data (Tata motors)



References

Github Repo

Wikipedia

Thanks!

